Using Google Colab to Run R Analyses of HINTS Data

Module Objectives

By the end of this module, you will be able to:

1. Use Google Colab to open, run, and save R notebooks.

2. Run and modify R code to conduct analyses of a public health dataset.

3. Interpret coefficients and odds ratios from regression model results and summarize key findings.

What is Google Colab?

Google Colab ("Colaboratory") is a free, browser-based coding environment. It runs Jupyter-style notebooks in the cloud. This means you can run R analyses without needing to have R installed on your computer. For this course, we'll be using Colab with the **R language** and a shared template notebook.

What you will need:

- A Google account (UF or personal)
- A reliable internet connection
- The course's Colab notebook file. A link to download this file is provided in Step 1 of the instructions below.

Important: Colab uses temporary machines. Each time you open a new session, previously installed R packages are cleared. The notebook we provide includes code at the top to install everything you need, so just make sure to run the notebook from top to bottom when you start.

Let's get started! Please follow the steps below. Screenshots are provided to guide you.

1. First, please download the course notebook here: Module 2 CoLab NotebookDownload Module 2 CoLab Notebook
2. Visit: https://colab.research.google.com/Links to an external site.
3. By default, Colab's language is set to Python, but we will be using R. In the top menu, click "Runtime" → "Change runtime type". Change the runtime type to R, then click save.

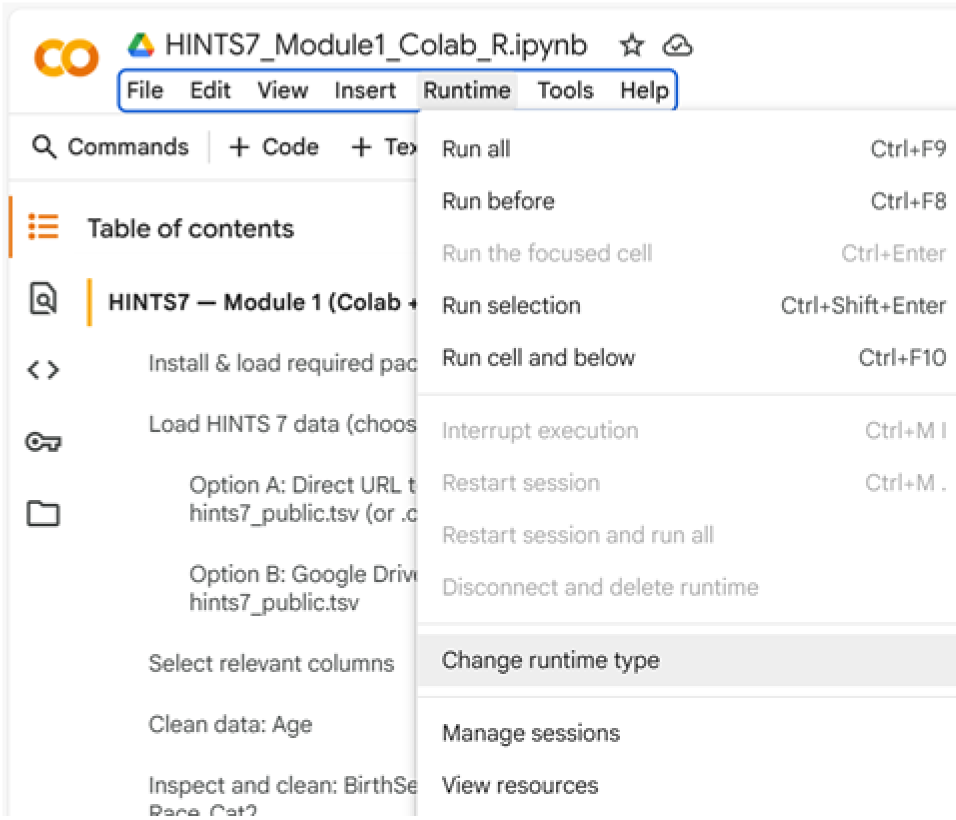
HINTS7_Module1_Colab_R.ipynb
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Table of contents
HINTS7 — Module 1 (Colab
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Option B: Google Driv hints7_public.tsv
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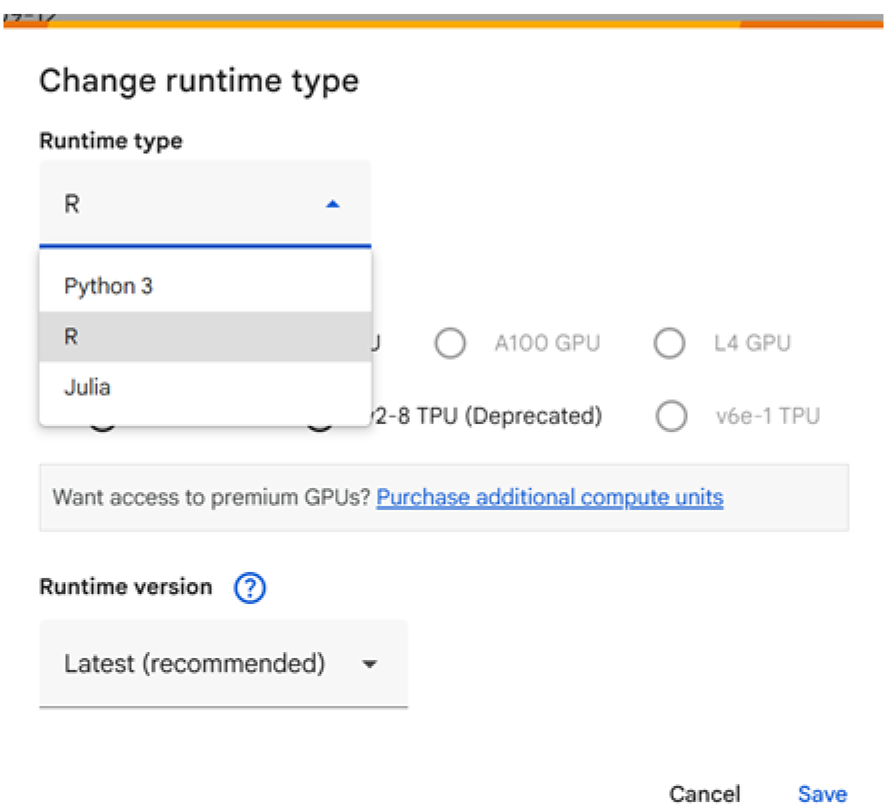
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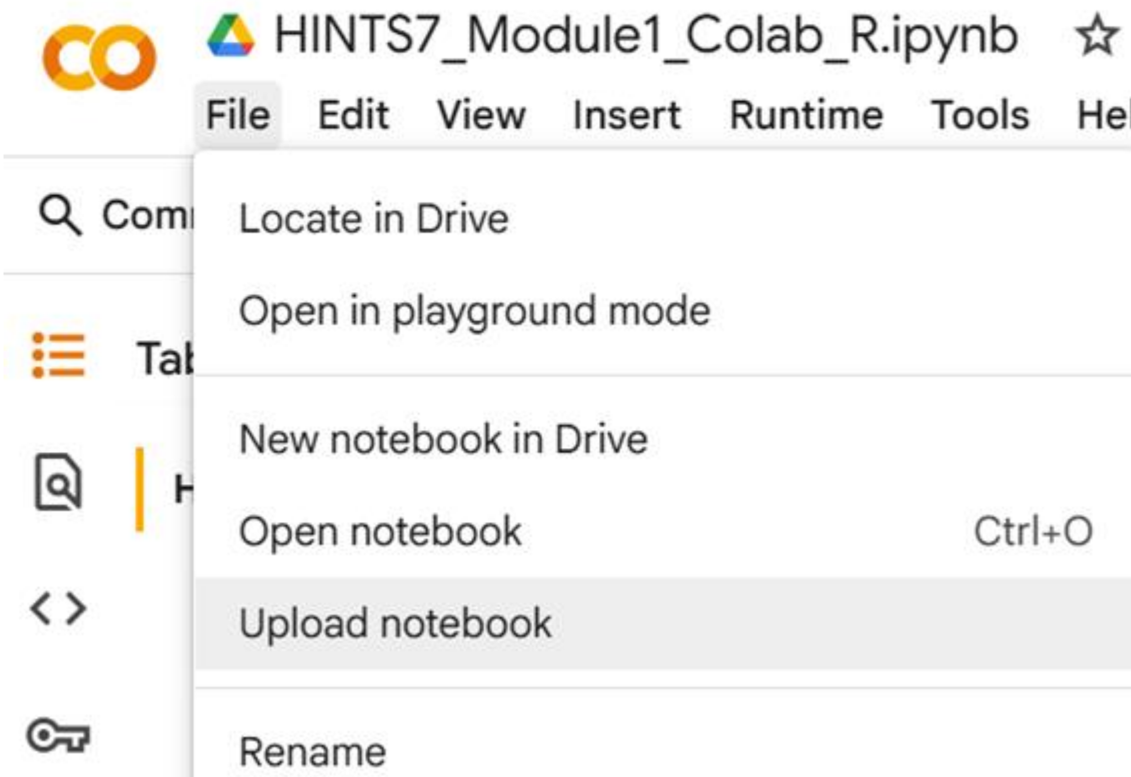
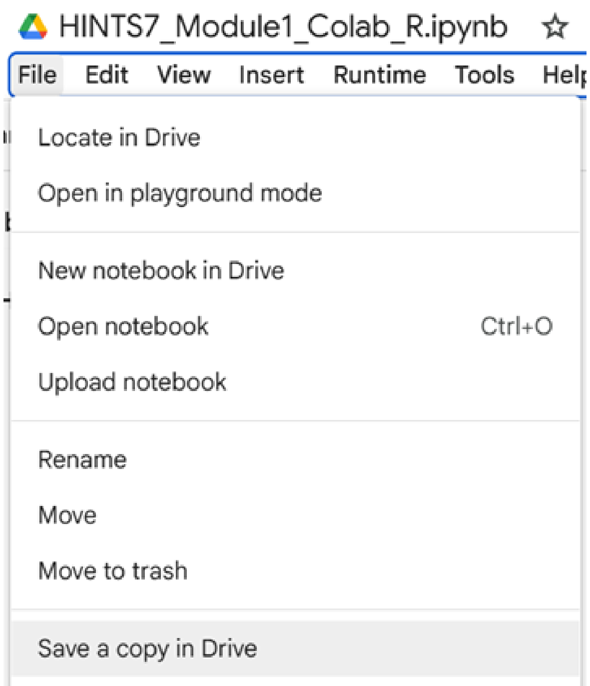
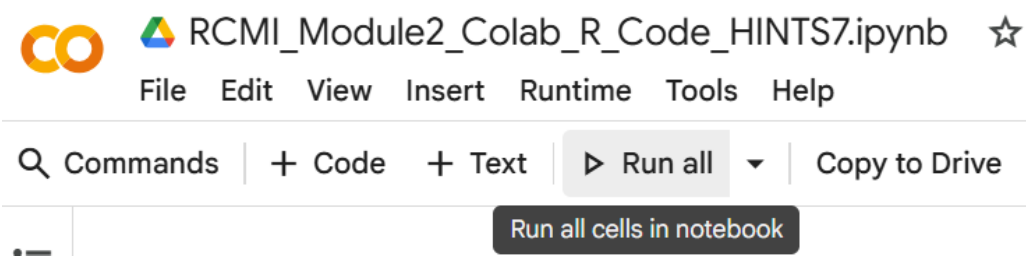
4. Next, in the top menu, click “File” → “Upload Notebook”, then upload the course notebook.



5. Go to “File” → “Save a copy in Drive” to make your own copy. You’ll edit and submit this copy.



6. Now, you can run the complete code in one go, by clicking the "Run all" button in the toolbar.



After running the code, you will see that output has been generated in response to three different research questions. You will choose one of these research questions to focus on in your assignments.

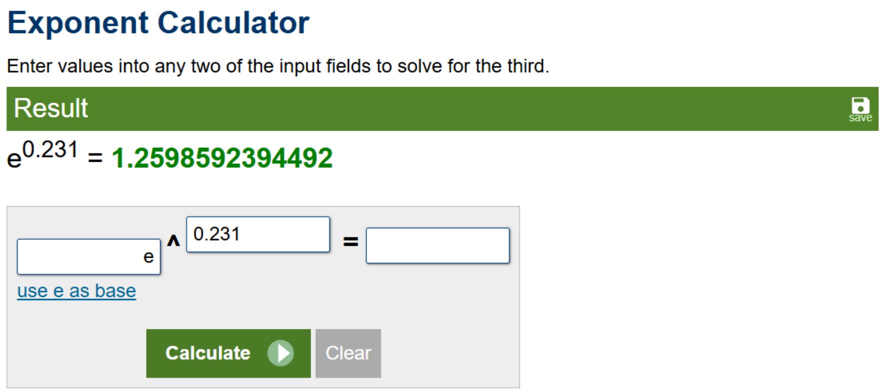
Take some time to read through the descriptions of the analyses for each research question and the tips on how to interpret the results. This will help you feel more confident and make it easier to decide which question you'd like to work on.

<u>How to Interpret the Coefficients</u>

Keep in mind how to interpret the coefficients for each type of model:

Research Question 1 (Ordinal logistic regression):
Coefficients are in log-odds. To interpret them as odds ratios, you need to exponentiate the coefficients (e.g., exp(0.231) = 1.26).

You can do this using a handheld calculator or an online calculator, such as Exponent Calculator. Use e as the base, and enter the coefficient, e.g., 0.231 as shown in the screenshot below:



Research Question 2 (Linear regression):
Coefficients are in the original scale of the outcome (patient–provider communication quality score). No exponentiation is needed.

Research Question 3 (Binary logistic regression):
Coefficients are in log-odds. To interpret as odds ratios, you need to exponentiate them. Please see the note above for Research Question 1 which provides guidance on exponentiating using an online calculator.

<u>Research Question 1: Does the use of wearable health technology correlate with frequency of healthcare visits or care coordination quality?</u>

Analysis:

Ordinal logistic regression for two outcomes:

Frequency of provider visits (None → 10+ times).

Care coordination quality (Never → Always).
Predictors included wearable use, age, sex, race/ethnicity, income, and education.

Results:

Wearable use was significantly associated with both outcomes:

+26% higher odds of more frequent visits (OR = 1.26, $p < .001$).

+17% higher odds of reporting better care coordination (OR = 1.17, $p = .02$).

Other predictors:

Age: older adults more likely to report frequent visits and better coordination (OR = 1.02 per year).

Sex: males less likely to visit frequently (OR = 0.71); not significant for coordination.

Race: Non-Hispanic Black and White reported higher odds for visits and coordination; Non-Hispanic Asian reported lower odds.

Education: higher education linked to better coordination.

Income: not significant.

Summary:

Wearable users had more frequent provider visits and reported better care coordination, independent of demographics, income, and education. Age, sex, race/ethnicity, and education also played important roles.

Research Question 2: How are symptoms of depression associated with reported quality of patient–provider communication?

Analysis:
Linear regression predicting patient–provider communication quality (PCC scale, continuous) from depression status, age, sex, race/ethnicity, income, and education.

Results:

Depression: respondents with depression scored about 3 points lower on the PCC scale ($p < .001$).

Age: older adults reported slightly better communication (+0.15 per year, $p < .001$).

Race/ethnicity:

Non-Hispanic Black respondents reported higher scores (+4.8, $p < .001$).

Non-Hispanic Asian (–4.0, $p = .01$) and multiracial (–4.2, $p = .02$) reported lower scores.

Sex: no significant difference.

Income and education: some contrasts significant but mixed and harder to interpret.

Model fit:
The model explained only ~3% of the variance (adjusted $R^2 = 0.032$).

Summary:
Depression was significantly linked to worse reported communication with providers. Age and race/ethnicity also mattered, but overall the model explained little of the variability in communication quality.

Research Question 3: How does exposure to online health information affect public trust in healthcare providers and systems?

Analysis:
Binary logistic regression predicting high trust in the healthcare system (1 = “a lot of trust”) from online health information use, online test result access, and demographic variables.

Results:

Online health information: not significantly associated with trust (OR = 1.0).

Online test results: not accessing results online was linked to 27% lower odds of high trust (OR = 0.73, $p < .001$).

Demographics:

Age: older adults more likely to report trust (OR = 1.02 per year, $p < .001$).

Sex: males more likely to report trust (OR = 1.17, $p = .006$).

Race: Non-Hispanic Asian respondents more likely to report trust (OR = 1.32, $p = .04$).

Education: less than high school associated with lower trust (OR = 0.71, $p = .02$).

Income: not significant.

Summary: General online health information use was unrelated to trust, but access to online test results was strongly linked to greater trust. Age, sex, race, and education also shaped trust, while income did not.